Geophysical Research Letters

RESEARCH LETTER
10.1029/2021GL092544

Key Points:

- Composite of subsidence, sea-level rise, and storm surge models
- Subsidence intensifies sea-level rise and increases inundation area up to 39%

Supporting Information:

Supporting Information may be found in the online version of this article.

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Citation:

Miller, M. M., & Shirzaei, M. (2021). Assessment of future flood hazards for southeastern Texas: Synthesizing subsidence, sea-level rise, and storm surge scenarios. *Geophysical Research Letters*, *48*, e2021GL092544. https://doi.org/10.1029/2021GL092544

Received 14 JAN 2021
Accepted 16 MAR 2021

Assessment of Future Flood Hazards for Southeastern Texas: Synthesizing Subsidence, Sea-Level Rise, and Storm Surge Scenarios

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Abstract Recent hurricanes highlight shortcomings of flood resilience plans in Texas that can worsen with climate change and rising seas. Combining vertical land motion (VLM) with sea-level rise (SLR) projections and storm surge scenarios for the years 2030, 2050, and 2100, we quantify the extent of flooding hazards. VLM rates are obtained from GNSS data and InSAR imagery from ALOS and Sentinel-1A/B satellites. VLM is resampled and projected on LIDAR topographic data, then multiple inundation and flooding scenarios are modeled. By the year 2100, over 76 km^2 are projected to subside below sea level. Subsidence increases the area of inundation over SLR alone by up to 39%. Under the worst-case composite scenario of an 8-m storm surge, subsidence, and the SLR RCP8.5, the total affected area is 1,156 km^2. These models enable communities to improve flood resiliency plans.

Plain Language Summary Flooding in southeastern Texas is likely to worsen as the sea-levels rise, the land subsides, and storms intensify. We model combined vertical land motion, sea-level rise, and storm surge scenarios for the years 2030, 2050, and 2100. Vertical land motion is obtained by combining data from global positioning stations and satellite radar interferometry. The subsidence rates are projected forward in time and resampled on a high-resolution topographic model. We then model the inundated area due to increasing sea levels, various storm surge heights, and land subsidence. Considering only land subsidence, 76 km^2 subsides below sea level by the year 2100. Compared to models of sea-level rise alone, subsidence increases the flooded area by up to 39%. Under the worst-case composite scenario, the total affected area is 1,156 km^2.

1. Introduction

Large portions of low lying (<10 m) coastal lands are subject to land subsidence (Shirzaei et al., 2021), which exacerbates flooding hazards due to sea-level rise (Shirzaei & Bürgmann, 2018) and affecting the vast population residing in these areas (Milliman & Haq, 1996). Southeastern Texas is subject to coastal land subsidence, including Houston, the fourth-most populous city in the U.S. (Miller & Shirzaei, 2019; Qu et al., 2015). Furthermore, the area has been ravaged by major Hurricanes, including Category-4 Hurricane Harvey. On August 25, 2017, the hurricane made landfall, then stalled for 3 days, causing a rare 9000-years extreme precipitation event as a tropical storm (Van Oldenborgh et al., 2017). The disaster took the lives of 80 people, displaced multitudes, and damaged >80,000 houses without flood insurance (Shultz et al., 2017), many of which were beyond the Federal Emergency Management Agency (FEMA) designated 500-years flood zone (Blessing et al., 2017).

The average global SLR rate of ∼3.3 mm/yr has been subject to an acceleration of 0.1 mm/yr^2 since 1993 (Cazenave et al., 2018). Sea-level rise (SLR) projections are based on the Representative Concentration Pathways (RCP) scenarios that consider a range of possible radiative forcings and indicate that the acceleration is likely to continue through the 21st century (Horton et al., 2018). Owing to changes in the global climate and sea level (Seddon et al., 2016), the frequency of severe flooding events has increased worldwide (Taylor et al., 2017). Climate change increases the probability of hurricane flood hazards (Marsooli et al., 2019) through hurricane intensification and sea-level rise (Woodruff et al., 2013), tropical cyclone precipitation rates (Liu et al., 2019), increased storm surge magnitudes (Lin et al., 2012), and compound flooding risk for many densely populated coastal cities (Hirabayashi et al., 2013; Wdowinski et al., 2020).

Urbanization augments flood risks by changes in land usage and cover (Day et al., 2007) and conversion to less permeable ground cover (Liscum, 2001). During Hurricane Harvey, models indicate urbanization exacerbated both the flood response and total rainfall, increasing the probability of an event of that severity by about 21 times (Zhang et al., 2018). Flooding severity can also increase due to land subsidence. Subsidence can lower flood control structures (Dixon et al., 2006) and alter floodplain boundaries and base flood drainage making them less effective (Wang et al., 2012). Miller and Shirzaei (2019) showed that 89% of the flooded area had been subsiding at a rate of 3 mm/yr or more before Hurricane Harvey.

Hurricane surge depends on topographic features and hurricane meteorological characteristics of radius, maximum wind, speed, and track angle, and acts as a protracted period wave (Rego & Li, 2010). Galveston Bay recorded high water marks nearing 6 m during 2008 Hurricane Ike (Irish et al., 2011). During Hurricane Harvey, the highest storm surge of 3.7 m was at the Aransas Wildlife Refuge near Corpus Christi (Sebastian et al., 2017), and remained 0.8 m above expected tides for ∼5 days in lower Galveston Bay (Valle-Levinson et al., 2020).

To prepare for a future in which the odds of extreme flooding double approximately every 5 years (Taherkhani et al., 2020), coastal cities are developing resilience strategies to avoid or reduce vulnerability to flooding (Aerts et al., 2014) and reexamining flood risk assessments (Bilskie & Hagen, 2018). This study advances understanding of future flooding and inundation extent by calculating and projecting vertical land motion (VLM), SLR, and storm surge scenarios for years 2030, 2050, and 2100. Vertical land motion is vertical displacement of the land surface referenced to the Earth's center of mass and determined with geodetic techniques (Wöppelmann & Marcos, 2016). Here, VLM is determined by combining InSAR and GNSS data and projected onto a high-resolution LIDAR digital elevation model (DEM) (Figure 1a). The subsidence models are then subjected to a combination of SLR scenarios for RCP2.6, RCP4.5, RCP6.0, and RCP8.5 projections and five simplified storm surge scenarios to identify areas subject to future inundation or flooding.

2. Methods and Data

We begin by examining the local sea level trends observed at the Galveston II, Pier 21, TX tide gauge (Holgate et al., 2013; PSMSL, 2020). The monthly mean measurements are referenced to a common reference datum, and a linear fit of the long-term monthly measurements at the tide gauge yields a relative mean sea level rate of 6.5 mm/yr (Figure 1a). For use as a high-resolution DEM as a model base, Texas Natural Resources Information System (TNRIS) supplies 50-cm resolution LIDAR data covering portions of the Texas upper coastal counties acquired in 2018 (Figure 1b).

InSAR and GNSS data are used to detect and monitor land subsidence due to natural and anthropogenic processes in the Houston-Galveston area (Buckley, 2003; Khan et al., 2014; Miller & Shirzaei, 2019; Qu et al., 2015). To obtain a high-resolution map of VLM, following earlier studies (Blackwell et al., 2020; Miller & Shirzaei, 2019; Shirzaei, 2013; Shirzaei & Bürgmann, 2018) we begin with performing an advanced multitemporal InSAR approach. The SAR data sets include 83 ascending L-band ALOS SAR images from seven overlapping frames from 2007 to 2011 (Table S1) and 108 ascending C-band Sentinel-1A SAR images from two overlapping frames from 2016 to 2019 (Table S2). For details on multitemporal interferometric processing of each SAR data set, see supporting information. Once these nine frames are processed and the LOS velocity maps obtained (Figures S1 and S2), the frames are mosaiced to generate two seamless maps of LOS velocities for the ALOS and Sentinel data sets. To this end, adjacent frames are aligned by identifying tie points within overlapping areas and calculating constant shifts using differences between mean LOS rates (Ojha et al., 2018). Next, we obtain horizontal velocities from GNSS stations (Blewitt et al., 2016), referred to International GNSS Service 2014 (IGS14) reference frame (Courtesy UNR). We apply a Kriging interpolation weighted with inverse distance (Trochu, 1993) to oversample Sentinel LOS velocities and GNSS horizontal velocities onto ALOS pixel locations, resulting in four observations per pixel with three unknowns, including velocities in East, North, and Up. Given the interpolated LOS velocities $\left\{y_{sa}, y_{aa}\right\}$ and variance $\left\{\sigma_{sa}^2, \sigma_{aa}^2\right\}$ for a given pixel, where subscripts indicate Sentinel-1 ascending (sa) and ALOS ascending (aa), we apply the following stochastic model:

$$y_{sa} = C_e^{sa} E + C_n^{sa} N + C_u^{sa} U + \varepsilon^{sa},$$

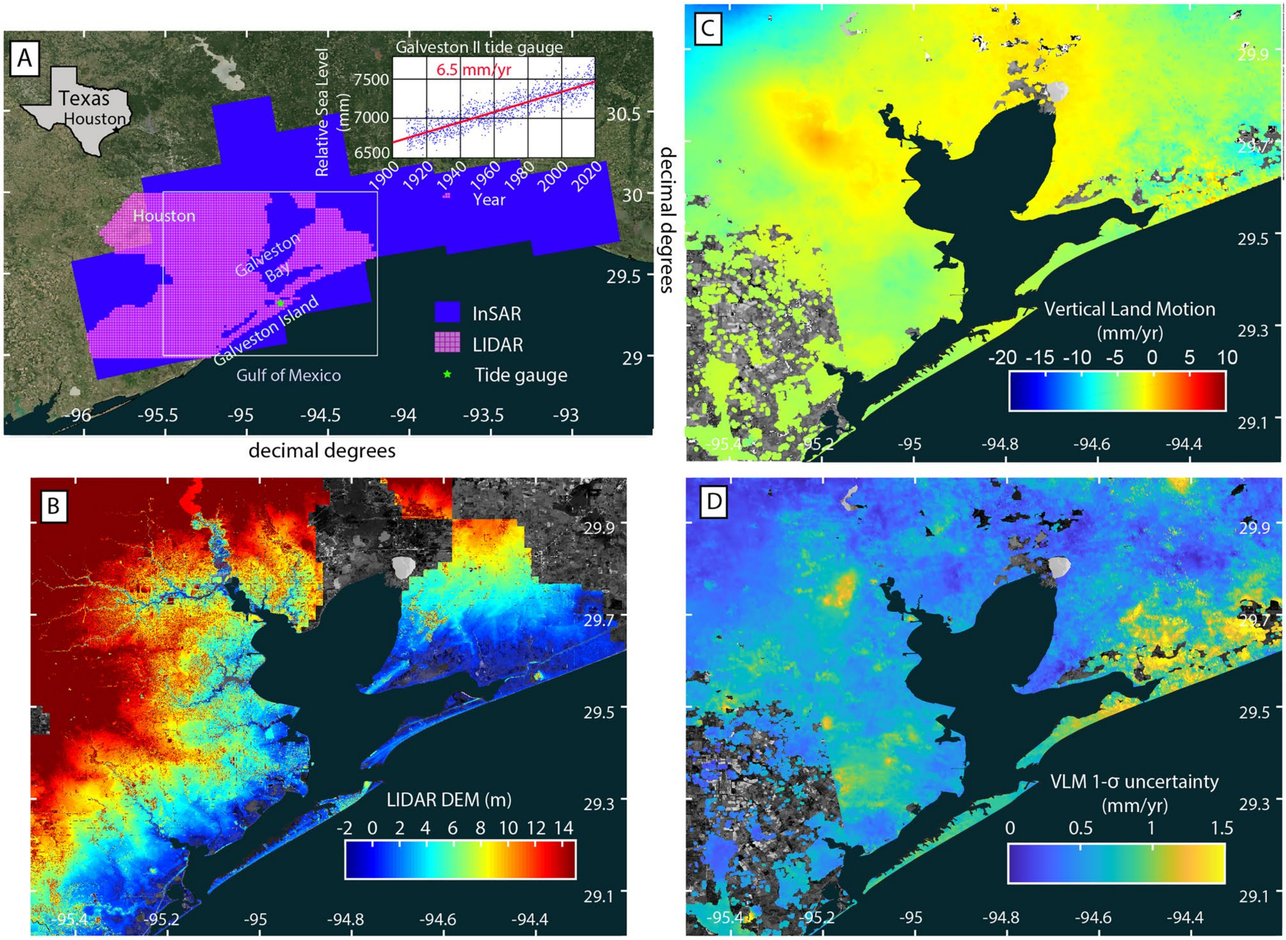

Figure 1. (a) Overview of InSAR and LIDAR data extents and relative sea level time series at Galveston II tide gauge, (b) LIDAR DEM. (c) Vertical land motion rate map derived from the combination of InSAR and GNSS data with respect to the IGS14 global reference frame and (d) Standard deviation of VLM.

$$y_{aa} = C_e^{aa} E + C_n^{aa} N + C_u^{aa} U + \varepsilon^{aa},$$

$$E_G = E + \varepsilon^e,$$

$$N_G = N + \varepsilon^n, \tag{1}$$

where E (east), N (north), and U (up) motions are unknowns, C represents the unit vectors projecting the 3D displacements onto the LOS (Hanssen, 2001), E_G and N_G are the interpolated GNSS velocities in E-W and N-S directions, and ε is the observations error with $N(0,\sigma)$ probability distribution (σ is the standard deviation). Note that in Equation 1, E and N motions are both unknown and observed. Thus, we use the unified least-squares framework, in which unknowns can be observed (see Chapter 12, Mikhail & Ackermann, 1982). The σ for the LOS velocities is considered 5 mm/yr and that of GNSS E-W and N-S components is provided by Nevada Geodetic Laboratory (Blewitt et al., 2016). The solution to Equation 1 is given by

$$X = \left(A^T P A\right)^{-1} A^T P L, \tag{2}$$

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where

$$X = \left(E\, N\, U\right)^T,$$

$$A = \begin{pmatrix} C_e^{sa} & C_n^{sa} & C_u^{sa} \\ C_e^{aa} & C_n^{aa} & C_u^{aa} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

$$P = \begin{pmatrix} \frac{1}{\left(\sigma^{sa}\right)^2} & 0 & 0 & 0 \\ 0 & \frac{1}{\left(\sigma^{aa}\right)^2} & 0 & 0 \\ 0 & 0 & \frac{1}{\left(\sigma^{e}\right)^2} & 0 \\ 0 & 0 & 0 & \frac{1}{\left(\sigma^{n}\right)^2} \end{pmatrix}. \quad (3)$$

The parameters' variance-covariance matrix is $Q_{XX} = \sigma_0^2\left(A^T PA\right)^{-1}$, where $\sigma_0^2 = \frac{r^T Pr}{n-u}$, $r = L - AX$, n is the number of equations, and u is the number of unknown. The diagonal elements of Q_{XX} are the variance of E, N, and U unknowns.

The distribution of E and N motions and the associated 1-sigma errors are shown in Figure S3, as well as comparisons of the combined land motion results with GNSS velocities. Here, we only focus on the maps of the vertical deformation rate concerning the IGS14 reference frame shown in Figure 1c and 1-sigma errors provided in Figure 1d. Subsidence rates during the study period of 2007–2019 reach 13.3 ± 1.5 mm/yr in northwest of Houston. Locations experiencing subsidence closer to the coast occur between Dickinson and Texas City, southwestern Galveston Island, and the Bolivar Peninsula. Some uplift occurs in the vicinity of Pasadena. These results agree with recent subsidence studies using GNSS (Wang et al., 2017) and InSAR (Qu et al., 2015). Residuals between land motion velocity solutions and GNSS velocities are shown in Figure S3 and Table S3.

In the Houston area, land subsidence stems from anthropogenic activities, in particular aquifer overdraft (Buckley, 2003; Coplin & Galloway, 1999; Galloway et al., 1998) and hydrocarbon production (Holzer & Bluntzer, 1984; Sharp & Hill, 1995), as well as natural processes of sediment compaction, tectonic processes (Kreitler, 1977) and Glacial Isostatic Adjustment (Peltier et al., 2018). However, the contribution from GIA is comparatively minor. Figure S4 shows the VLM predicted due to GIA following the ICE-6G_D model (Peltier et al., 2015). The study area is characterized by <1 mm/yr of GIA subsidence, which is in the range of observation error. Furthermore, due to the lack of localized features with a fast subsidence rate, we rule out the contribution from the contemporary depletion of aquifer and reservoirs (Shirzaei et al., 2021). Residual aquifer compaction from groundwater withdrawal and natural sediment compaction are the drivers of the observed VLM in the Houston-Galveston area. Uplift in the study area may be attributed to poroelastic rebound from groundwater recovery, salt diapir motion, or related movements along faults, which is an opportunity for future study.

Next, we resample the VLM rates on a 1 × 1 m grid, downsampled from the LIDAR DEM (Strategic Mapping Program, 2018). Assuming a linear rate for the VLM during the 21st century (Shirzaei et al., 2021), we adjust the elevation model to incorporate subsidence projections. The subsidence affected elevation models are then subjected to multiple SLR and storm surge scenarios by subtracting the height of the transformed DEM from the SLR projection height. SLR projections are based on a range of possible radiative forcing values related to greenhouse gas concentrations from the Representative Concentration Pathways (RCP) adopted

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by the Intergovernmental Panel on Climate Change (IPCC). Using projections up to the year 2100, we examine 2-sigma confidence ranges for RCP2.6, RCP4.5, RCP6.0, and RCP8.5 (Table S4) (Horton et al., 2018). Five simplified storm surge scenarios (0.5, 1, 2, 4, and 8 m) are selected based on the ranges from complex coastline modeling of various cyclone landfall locations and intensities. However, individual storm dynamics result in a variable, nonuniform storm tide along the coast (Bass et al., 2018).

3. Future Flooding Hazards

The multiple combinations of subsidence with SLR and storm surges of different severities predict a range of flood and inundation extents that can inform future flood resilience plans (Table 1). Beginning with coastal inundation models from subsidence alone (Figure 2), the total area affected reaches 29.3 km^2 by the year 2030, 39.7 km^2 by the year 2050, 76.1 km^2 by the year 2100, and affects marshlands and bayous primarily. Next, we model the area affected by SLR alone for the mean RCP2.6, RCP4.5, RCP6.0, and RCP8.5 scenario values and the 2-sigma range in the supplement (Table S4 and Figure S5). We do not distinguish between hydrologically connected or unconnected areas, or between natural or urban structures. We proceed using

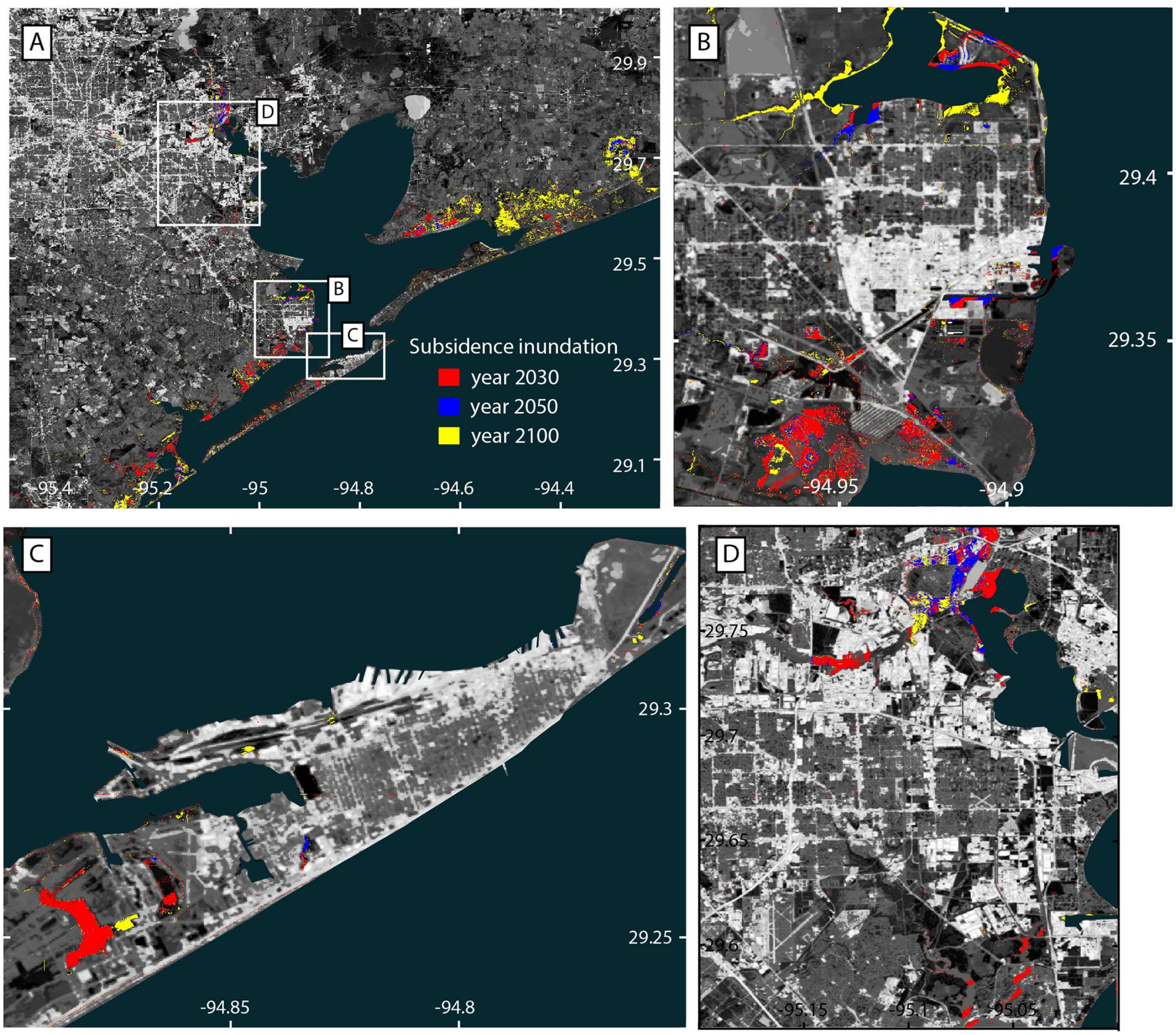

Figure 2. Subsidence inundation by years 2030, 2050, and 2100 for (a) the Houston-Galveston region, including (b) Texas City, (c) Galveston Island, and (d) southeastern Houston. The progressive color code means areas inundated by later years include the areas inundated in earlier years.

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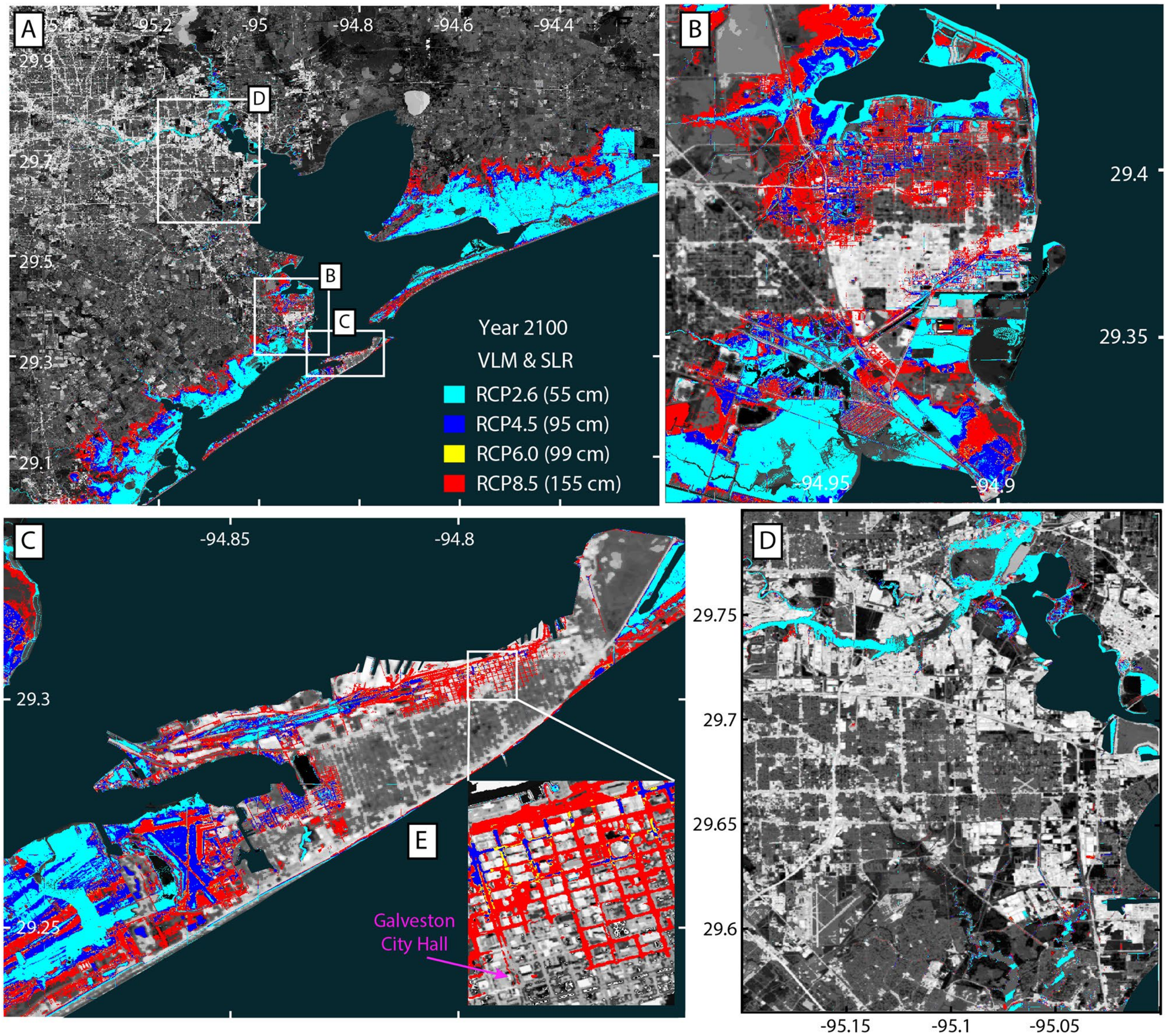

Figure 3. Year 2100 combined VLM and SLR submersion under RCP scenarios: RCP2.6 (cyan), RCP4.5 (blue), RCP6.0 (yellow), and RCP8.5 (red) for (a) the Houston-Galveston region, including (b) Texas City, (c) Galveston Island, and (d) southeastern Houston, and (e) downtown Galveston. The progressive color code means areas with higher SLR values include all the areas represented by colors for lower SLR values. RCP, Representative Concentration Pathways; SLR, sea-level rise; VLM, vertical land motion.

the mean RCP estimates for modeling VLM and SLR combined. By 2100, the extent of the inundation area from RCP2.6 to RCP8.5 is 221.63–361.88 km^2, respectively (Figure 3 and Table 1); the 2030 result is shown in Figure S6 and 2050 result in Figure S7. VLM increases the area of inundation over SLR alone by 9% for the year 2030 and ~39% by the year 2050 (Figure 5a). By 2100, the increase attributable to the addition of VLM varies by RCP scenario from 39% to 8% due to the increase in SLR and the topographic gradient. For all scenarios, portions of urban areas of Galveston and Texas City are at risk, though Houston remains mostly unaffected. As expected, storm surge flooding compounded with inundation from subsidence and sea-level rise affects more significant areas over time (Figure S8). By the year 2100, under the RCP8.5 scenario, much of Texas City and Galveston are overwhelmed by only a 0.5-m storm surge (Figure 4). As the storm surge height increases, the Houston metropolitan area is increasingly inundated (Figure 5b). Under the worst-case composite scenario of an 8-m storm surge, subsidence, and the RCP8.5 SLR scenario, the total affected area is 1156.68 km^2. Although this scenario is extreme, the exercise allows for exploring a *worst-case* scenario and gives perspective on potential flooding patterns. Note that the future sea level likely resides

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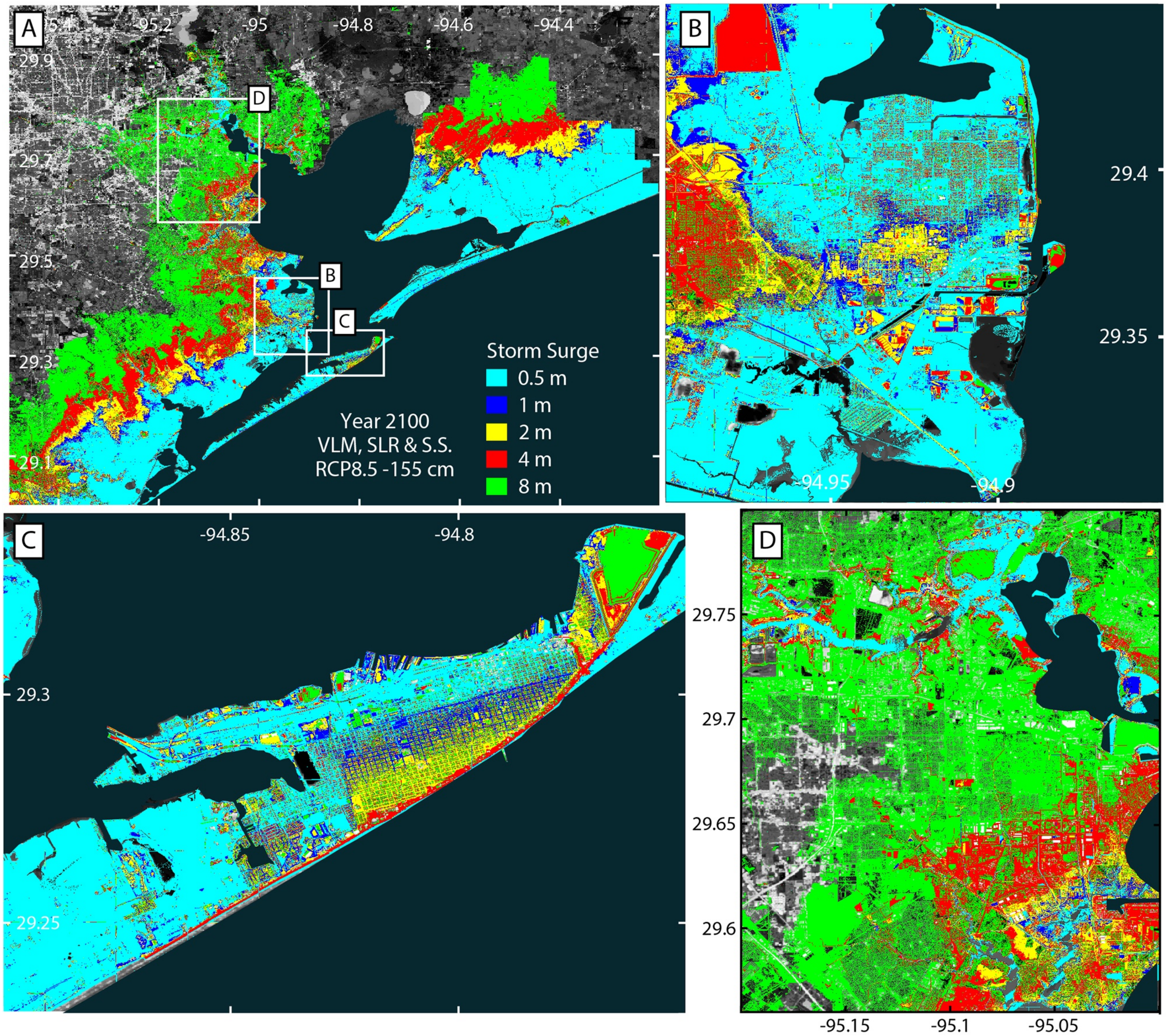

Figure 4. Inundation via subsidence, SLR RCP8.5 by year 2100, and storm surge flooding for (a) the Houston-Galveston region, including (b) Texas City, (c) Galveston Island, and (d) southeastern Houston. The progressive color code means areas with higher surge values includes all the areas represented by colors for smaller surge values; e.g., the green area includes green and all the areas represented by smaller surges. RCP, Representative Concentration Pathways; SLR, sea-level rise.

somewhere between the levels predicted by different RCPs scenarios. Thus, an ensemble of inundation area estimates obtained for different scenarios constitutes the uncertainty range of future inundation hazards.

4. Discussion

Houston and Galveston are vulnerable to the combined effects of global climate change, mainly nuisance flooding, long-term SLR, and intensified hurricanes. The Texas Demographic Center produces projections of county populations, which places estimates of combined Harris and Galveston counties at 8.5 million people by 2050 (Potter & Hoque, 2014). Careful consideration must be given to land usage, freshwater resources, oil and gas extraction, and urbanization during this period of population growth. Since the 1900s, groundwater has been the primary municipal, commercial, and industrial source of water for Houston and Galveston, but regulatory plans to reduce groundwater reliance are being gradually implemented (Shah

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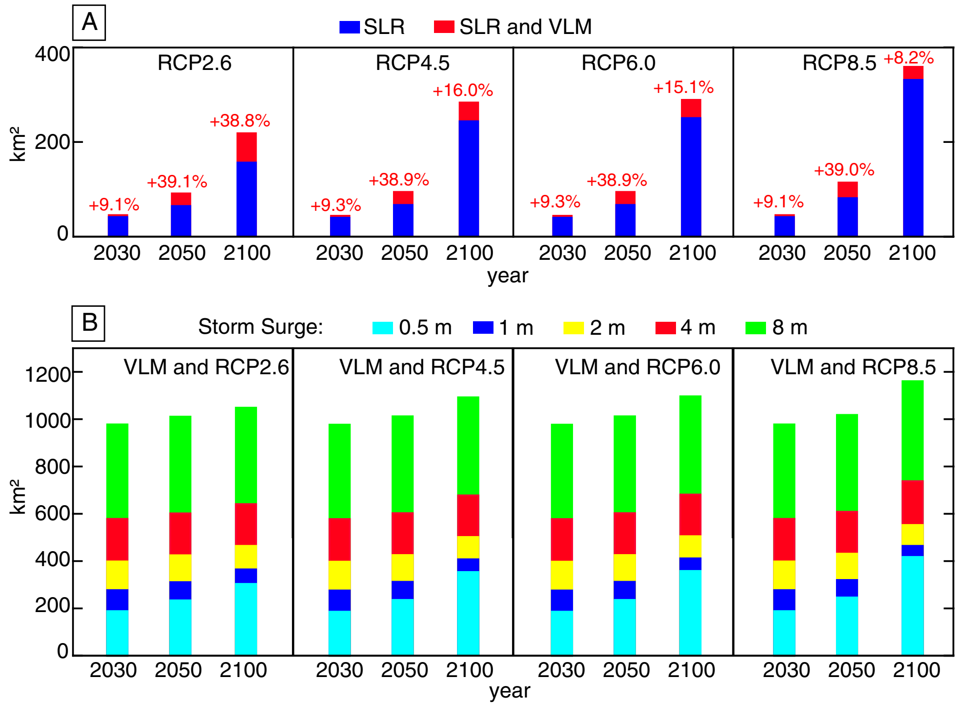

Figure 5. (a) Areas of inundation from SLR (blue) and SLR and VLM (blue + red) with percentage increases for each scenario. (b) Areas affected by flooding and inundation with SLR, VLM, and progressive storm surge heights (cumulative). SLR, sea-level rise; VLM, vertical land motion.

et al., 2018). Depending on the exponential decay of displacement represented by the compaction time constant, a linear trend may overestimate the area of inundation. However, we estimate residual compaction due to past pumping can continue for 280–400 years, and in this case, overestimation would be minimal (Miller & Shirzaei, 2019). In fact, subsidence has the potential to accelerate in the event of a return to aquifer overexploitation. With increased aquifer overdraft, the assumption of a linear trend for subsidence will underestimate the inundation area. Additionally, SLR, coastal land subsidence, and increasing water demands exacerbate the threat of seawater intrusion into freshwater aquifers (Jasechko et al., 2020), which is relied upon by the growing population.

The probability of Harvey-like rainfall, estimated to be ∼1% during the period 1981–2000, rises to 18% for the period 2081–2100 (Emanuel, 2017). Recently in 2020, an active season brought three tropical storms close to our study area. Although we choose a simplified uniform surge height for our model, storm surge dynamics are dependent on tropical cyclone characteristics along the complex coastline and future works would do well to explore storm surge variability (Bass et al., 2018).

More accurate hydrologic models incorporating subsidence and sea-level rise can aid in remapping flood hazard zones and improve flood resilience. By 2100, a population of 300,000–500,000 people may face difficult decisions to protect, accommodate, or retreat from SLR, of which there are considerations for environmental, political, social, demographic, and economic factors (Hauer et al., 2020). The population of Northern America residing in flood plains is projected to increase from 4.2 million (year 2000) to about 8.0 million by 2060 (Neumann et al., 2015). Coastal populations worldwide are expected to grow greater than 300% by 2070 and property affected by flash flooding will value ∼9% of the projected global GDP (Aerts et al., 2014). Combining remote sensing techniques with modeling will provide detailed vertical land motion observations and high-resolution inundation models useful for planning. The techniques implemented in this study can be used to evaluate other cities, inform policy decisions, improve hazard risk assessments, and flood resilience strategies.

Table 1

Modeled Inundation Area for Subsidence, Sea-Level Rise (SLR), Storm Surge (S.S.), and Combined Scenarios Through the 21st Century

		Year					
		2030		2050		2100	
		Area (km^2)					
Subsidence			29.34		36.69		76.08
		Year					
		2030		2050		2100	
		SLR height (cm)	Area (km^2)	SLR height (cm)	Area (km^2)	SLR height (cm)	Area (km^2)
Sea-level rise (SLR)	RCP2.6	13	44.68	25	68.10	55	159.73
	RCP4.5	12	43.27	26	70.45	95	246.94
	RCP6.0	12	43.27	26	70.45	99	253.73
	RCP8.5	13	44.67	32	84.66	155	334.34
Subsidence and SLR	RCP2.6	13	48.76	25	94.71	55	221.63
	RCP4.5	12	47.29	26	97.82	95	286.52
	RCP6.0	12	47.29	26	97.82	99	292.13
	RCP8.5	13	48.76	32	117.64	155	361.88
		Year					
		2030		2050		2100	
	SS height (m)	Area (km^2)					
Subsidence and storm surge (S.S.)	0.5		153.59		174.55		212.43
	1		257.69		268.51		293.54
	2		384.79		392.19		409.36
	4		567.57		572.87		586.13
	8		964.57		970.54		984.95
		SLR height: 13 cm		SLR height: 25 cm		SLR height: 55 cm	
Subsidence, SLR RCP2.6, and S.S.	0.5		188.62		227.48		300.48
	1		277.43		304.18		361.88
	2		398.78		417.70		461.15
	4		578.51		594.82		637.41
	8		978.02		996.00		1044.72
		SLR height: 12 cm		SLR height: 26 cm		SLR height: 95 cm	
Subsidence, SLR RCP4.5, and S.S.	0.5		186.23		229.26		350.65
	1		275.97		305.54		404.42
	2		397.74		418.67		498.59
	4		577.69		595.75		674.27
	8		977.00		997.04		1088.51
		SLR height: 12 cm		SLR height: 26 cm		SLR height: 99 cm	
Subsidence, SLR RCP6.0, and S.S.	0.5		186.23		229.26		355.21
	1		275.97		305.54		408.37
	2		397.74		418.67		502.02
	4		577.69		595.75		677.98
	8		977.00		997.04		1092.81

Table 1
Continued

		SLR height: 13 cm	SLR height: 32 cm	SLR height: 155 cm
Subsidence, SLR RCP8.5, and S.S.	0.5	188.62	239.46	414.31
	1	277.43	313.39	461.15
	2	398.78	424.42	549.40
	4	578.51	601.37	734.29
	8	978.02	1003.22	1156.68

Note: The model resolution is 1 m × 1 m, and total area values are aggregate and not contiguous.

Data Availability Statement

LIDAR data are available from TNRIS (https://data.tnris.org/collection/b5bd2b96-8ba5-4dc6-ba88-d88133eb6643). Tide gauge data obtained from Permanent Service for Mean Sea Level (https://www.psmsl.org/data/obtaining/stations/161.php). Sentinel-1 data obtained from European Space Agency (https://sentinels.copernicus.eu/web/sentinel/sentinel-data-access) and ALOS data from (https://earth.jaxa.jp/en.html).

Acknowledgments
We thank Cathleen E. Jones for offering improvements to the manuscript. M. Shirzaei was supported by the U.S. National Aeronautics and Space Administration Grant 80NSSC170567.

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